Requirements Gathering

Definition: A stakeholder is any individual, group, or organization that is affected by or can affect the outcome of this system. Identifying stakeholders early ensures the system addresses real needs and avoids conflicts during development.

Stakeholder Analysis

1. Mental Health Professionals	Psychiatrists, Psychologists, Counselors — Primary validators of system outputs
Key Needs	• Accurate risk predictions with clinical-level reliability • Explainable AI outputs (not a black box) to support clinical decisions • Confidence scores and risk level breakdown (Low / Medium / High) • Easy integration with existing patient management workflows
Main Concerns	• Over-reliance on AI replacing professional judgment • False positives causing unnecessary alarm for patients • Liability and accountability if the system misses a case • Data privacy and compliance with medical ethics (HIPAA / local regulations)

2. End Users (Individuals /Social Media Users)	People who voluntarily submit their posts for mental health screening
Key Needs	• Simple, non-clinical language in all outputs • Complete privacy — data must not be stored or shared without consent • Helpful resources and next steps (not just a risk score) • Fast and accessible interface (mobile-friendly)
Main Concerns	• Fear of stigma if identified as at-risk • Misunderstanding of results — treating AI output as medical diagnosis • Trust in the system's fairness and accuracy • Uncertainty about how their data is used

3. Research & Academic Institutions	Universities, labs, and NLP/ML researchers studying mental health AI
Key Needs	• Reproducible methodology and publicly documented pipeline • Access to model performance metrics and ablation studies • Ability to extend or benchmark against the system • Ethical AI practices and bias mitigation documentation
Main Concerns	• Dataset bias affecting generalizability of results • Lack of transparency in deep learning decisions • Model performance on Arabic vs. English content • Reproducibility if datasets are not publicly available

4. System Administrators / MLOps Team	Technical team responsible for deploying, monitoring, and maintaining the system
Key Needs	• Robust MLflow experiment tracking and model versioning • Automated monitoring for data drift and model degradation • Clear REST API documentation for integration • Scalable deployment architecture on Azure ML
Main Concerns	• Model drift over time as social media language evolves • High computational cost of BERT inference at scale • Security vulnerabilities in the API endpoints • Downtime or failure in production deployment

5. University Supervisors & Evaluation Committee	Academic supervisors and graduation project reviewers
Key Needs	• Clear project scope, objectives, and deliverables • Demonstrated use of ML + DL + NLP techniques • Well-documented explainability and model evaluation • Professional UI/UX with a working demo
Main Concerns	• Scope creep — team attempting too much within one semester • Ethical handling of sensitive mental health data • Code quality, documentation, and reproducibility • Meaningful contribution beyond basic classification

User Stories & Use Cases

- **Text-Based Risk Analysis**

  As a social media user, I want to paste or type my recent posts and receive a mental health risk assessment, so that I can identify whether I might be experiencing depression or stress and seek help early.

  - User can input text of up to 5,000 characters
  - System returns a risk level (Low / Medium / High) within 5 seconds
  - Output includes a confidence score and highlighted risk-related keywords
  - A disclaimer is shown clarifying this is NOT a medical diagnosis

- **Explainable AI Output**

  As a mental health professional, I want to see which specific words or phrases contributed to the risk prediction, so that I can validate the AI's reasoning and use it to support my clinical assessment.

  - SHAP values or attention weights are visualized per word/token
  - Top 10 contributing features are displayed in a ranked list
  - Professional can toggle between 'Simple View' and 'Technical View'
  - Explanations are downloadable as a PDF report

- **Behavioral Pattern Analysis**

  As a system user, I want to input multiple posts over time and track changes in my mental health risk score, so that I can monitor trends in my emotional well-being and detect worsening patterns early.

- User can submit a batch of timestamped posts
- System shows a time-series chart of risk score evolution
- System flags significant increases in risk score (>20% change)
- Posting frequency analysis is included in the report

- **Admin Model Monitoring Dashboard**

  As a system administrator, I want to monitor model performance and detect data drift in real time, so that I can ensure the system remains accurate and reliable in production without manual checks.

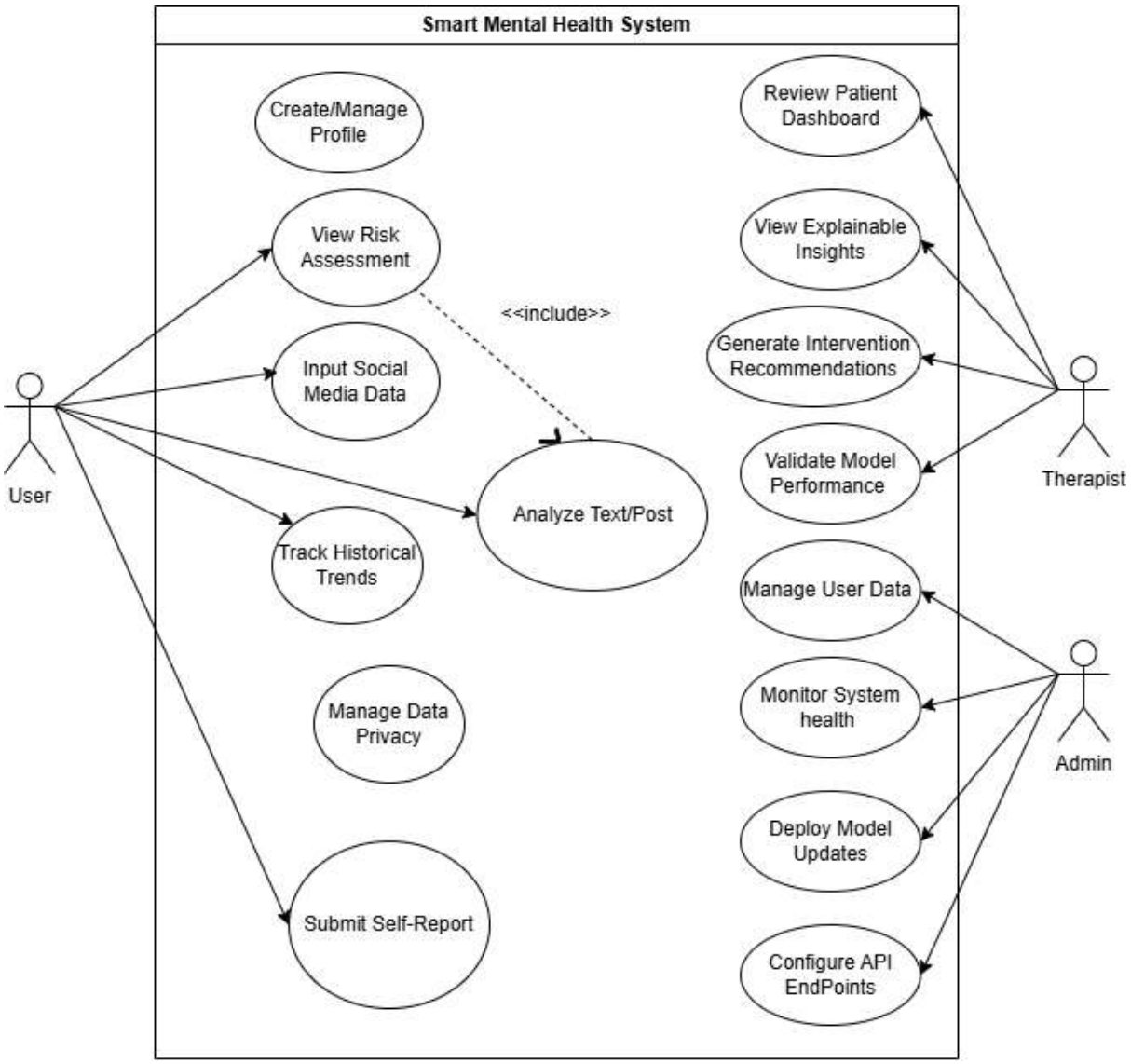
  - Dashboard shows live accuracy, F1-score, and AUC-ROC metrics
  - Drift detection alerts are triggered when distribution shifts by >15%
  - Admin can trigger model retraining from the dashboard
  - All experiments are logged in MLflow with version history

- **Anonymous & Privacy-Safe Analysis**

  As a privacy-conscious user, I want to use the system without my data being stored or linked to my identity, so that I can get mental health insights without fear of data misuse or privacy breaches.

  - No user account is required to use the system
  - Input text is not stored after analysis is complete
  - A privacy policy is displayed before analysis begins
  - System processes all data locally or with encrypted transit (HTTPS)

Use Case Diagram



Functional requirements

1. Text-Based Risk Analysis

- **Text Input:** The system shall allow users to input or paste text up to 5,000 characters for analysis.
- **Risk Classification:** The system shall categorize the mental health risk into three levels: Low, Medium, or High.
- **Scoring and Keywords:** The system shall provide a confidence score and highlight specific keywords related to the risk assessment.
- **Disclaimer Display:** The system shall display a mandatory disclaimer clarifying that the output is not a medical diagnosis.

2. Explainable AI (XAI) Output

- **Feature Visualization:** The system shall visualize SHAP values or attention weights for each word or token to explain the AI's reasoning.
- **Ranked Contributors:** The system shall display a ranked list of the top 10 features (words/phrases) that contributed to the prediction.
- **View Toggle:** The system shall allow professionals to toggle the interface between a 'Simple View' and a 'Technical View'.
- **PDF Export:** The system shall generate and allow the download of explanation reports in PDF format.

3. Behavioral Pattern Analysis

- **Batch Processing:** The system shall support the submission of multiple timestamped posts in a single batch.
- **Trend Visualization:** The system shall generate a time-series chart illustrating the evolution of the user's risk scores over time.
- **Risk Alerting:** The system shall automatically flag and alert the user to significant risk score increases exceeding 20%.
- **Frequency Reporting:** The system shall include a posting frequency analysis within the behavioral report.

4. Admin Model Monitoring Dashboard

- **Real-time Metrics:** The system shall provide a dashboard to monitor live performance metrics, including accuracy, F1-score, and AUC-ROC.
- **Drift Alerts:** The system shall trigger automated alerts when data drift or distribution shifts exceed 15%.

- **Retraining Control:** The system shall provide an administrative interface to trigger manual or automated model retraining.
- **Experiment Logging:** The system shall log all experiments and model versions using MLflow.

5. Privacy & Access

- **Anonymous Processing:** The system shall allow users to perform analyses without the requirement of creating a user account.
- **Data Volatility:** The system shall ensure that input text is not stored after the analysis is complete for privacy-conscious users.
- **Policy Transparency:** The system shall display a privacy policy to the user before any analysis begins.

Non-Functional requirements

Performance & Scalability

- **Response Time:** Risk assessments must be returned to the user within 5 seconds of input.
- **Scalability:** The system must use a scalable deployment architecture (such as Azure ML) to handle the computational costs of BERT inference.

Privacy & Ethics

- **Anonymity:** The system must allow analysis without requiring a user account or storing input text after the session ends.
- **Security:** All data must be processed with encrypted transit using HTTPS.
- **Compliance:** The system must adhere to data privacy standards and medical ethics (such as HIPAA or local regulations).
- **Transparency:** A clear disclaimer must be displayed stating that the system provides screening, not a formal medical diagnosis.

Usability & Reliability

- **Accessibility:** The interface must be mobile-friendly and use simple, non-clinical language for general users.
- **Reliability:** The system must maintain high availability, avoiding downtime or failures in the production environment.
- **Explainability:** AI outputs must be "explainable" rather than a "black box" to ensure clinical-level reliability for professionals.